LAS 6292: Data Collection and Management

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Preamble

This book is a manual for students in LAS 6292: Data Collection & Management. The course is designed for graduate students from any discipline – social sciences, humanities, biophysical sciences – and at all stages of their graduate program. It is an introduction to methods for collecting, organizing, managing, and visualizing both qualitative and quantitative data. Students will gain hands-on experience with best practices and tools.

Describe the different types of research data & the research data life-cycle -

Explain the need for and benefits of data management and sharing

Describe and implement best practices for data collection, storage, management, and sharing

Find, download, and analyze publicly available data from repositories

Carry out simple and reproducible data corrections and dataset organization

Describe public policies and agency requirements for data management and sharing

Articulate the major legal & ethical issues regarding the data collection, use, and storage

Create and Implement Data Management Plans in funder-specific formats

Identify and properly use tools for more efficient and secure data collection in the field

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Citation

This guide is a Quarto Book hosted on the Bruna Lab's Github site. Please suggest edits to the text, missing resources, or make suggestions for improvement either by pull request or by posting an issue on the course book's repository. For more information and tutorials for doing so, see **?@sec-manual**.

1. LAS 6292: Data Collection & Management

Spring Semester 2026

LAS6292 TCD Research Methods: Data Collection & Management Spring 2026

2. Instructor, Course Sessions, & Office Hours

Instructor: Dr. Emilio M. Bruna | embruna@ufl.edu | (352) 846-0634

Course Sessions: Friday 12:50-3:50 PM (Periods 6-8) in Grinter Hall 376

Office Hours: Tuesday and Wednesday from 10:30-11:45 am in person (380 Grinter Hall) or online (link provided on course Canvas page). You may sign up for a specific time slot here: https://embruna.youcanbook.me. Please let me know if you can't make these times and we will make alternative arrangements.

3. Course Description, Objectives, and Learning Outcomes

Courses taught as LAS 6292 are an introduction to field research methods for studies focused on natural resource use and management by local populations in tropical regions. They emphasize participatory approaches and integration of natural and social science tools.

This session of LAS 6292 is a practical introduction to ***Data Collection & Management***: the methods, tools, and best practices for collecting, organizing, managing, and visualizing qualitative and quantitative data. It is designed for graduate students from all disciplines at any stage of their program. At the conclusion of the course students will be able to*:*

1. Describe the different types of research data;
2. Explain the need for and benefits of data management and sharing;
3. Describe and implement best practices for the collection, storage, management, archiving, and sharing of research data;
4. Find, download, and analyze publicly available data from repositories;
5. Carry out simple and reproducible data corrections and data set organization;
6. Describe public policies and agency requirements for data management and sharing;
7. Articulate the major legal and ethical considerations regarding data collection, use, and storage (e.g., privacy/human subjects, intellectual property, international law);
8. Create and Implement Data Management Plan in funder-specific formats;
9. Identify and properly use tools for more efficient and secure data collection in the field.

4. Course Format, Prerequisites, & Credits

Format: In-person, **Credits:** 3, **Prerequisites:** None

Concentrations & Certificates: Doctoral Students can apply this course towards the requirements for the Certificate in Tropical Conservation & Development. See https://uftcd.org/academics/certificate/curriculum/ for more information.

5. Textbooks, Learning Materials, and Supply Fees

Students are not required to purchase any textbooks or course materials. All class materials & assignments will be made available on the course website. Some assigned readings may come from the *New York Times* or *Wall Street Journal* and have dynamic multimedia data visualizations and video that can't be appreciated in the posted .pdf format. *Students in this class should sign up for free online access to both the NYT and WSJ* by following the instructions at the UF Libraries Website.

Materials and Fees: None.

6. Course Schedule, Assignments & Grades

The contents of the course, including subjects covered in each course session and assigned readings, are subject to change based on student needs and input, current events, and technical advances related to the course subject matter. Changes will be announced via Canvas and an updated syllabus will posted.

Class Calendar

Week	Dates	Topic
1	1/16	‘Data’ across disciplines and the Research Data Life Cycle
2	1/23	File Formats, Naming Conventions, Data Storage & Security
3	1/30	Structure & Format of Data & Datasets
4	2/6	Reproducible Data (Re)organization
5	2/13	QA/AC 1: Data Entry & Validation
6	2/20	QA/QC 2: Correction & Synthesis with Open Refine
7	2/27	QA/QC 3: Visualizing Data (to Find Mistakes)
8	3/6	Metadata & Codebooks; Data Sharing, Reuse, & Archives
9	3/13	Data Management Plans
10	3/20	No Class - Spring Break
11	3/27	Efficient Data Collection
12	4/3	Transcription & Translation
13	4/10	Paperless Data Collection
14	4/17	Automated Data Extraction
15	4/24	No Class: Reading Days
Finals Week	4/30	Deadline for Final Project Submission: 10 AM

Final Course Grades & Grading Policy

Course grading is consistent with UF grading policies. **The final course grade is based on the assignments below.**. Most of the in-class assignments involve hands-on practice with data collection or manipulation. In some weeks, however, assignment will be the submission of questions for group discussion or brief reflection on the issues from the readings. Note also that most in-class assignments are designed to be completed during the class session, but to ensure students master the concepts rather than rush through them *they can be submitted anytime before 9 am the day of the following class session.*Late assignments will lose 10 pts. Finally, **there is no Final Exam in this course, and the Individual Project is due on the scheduled date of the final exam.**

Assignments

Assignment	Points (%)	Due
Weekly in-class exercises	350 (35%)	Following Friday at 9 am
Data Management Plan	150 (15%)	27 March
Individual Data Cleanup Project	500 (50%)	30 April
TOTAL	**1000 (100%)**	

Grading Scale

Course grades will be based on the percentage of total points earned and will be assigned with the following scale: **A:** >93% **A-:** 92-90% **B+:** 89-87% **B:** 86-83% **B-:** 82-80% **C+:** 79-77% **C:** 76-73% **C-:** 72-70%. **D+:** 69-67% **D-:** 62-60% **E:** 59% and below. *For information on how UF assigns grade points, visit:*
https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/

7. Required Reading & Preparation for Class Sessions

Please review the assigned material *before* class. *All items are posted on the Canvas site.*

Week 1: Course Introduction

No assigned reading

Week 2: File Formats, Naming Conventions, Data Storage, and Data Security.

Read:

1. Jan Čurn. 2014. How a bug in Dropbox permanently deleted my 8000 photos.
   [read online] [download pdf]

2. Panzarino, M. 2012. How Pixar's Toy Story 2 was deleted twice, once by technology and again for its own good. TNW. [read online] [[download pdf]]

3. Hart EM et al. (2016) Ten Simple Rules for Digital Data Storage. PLoS Comput Biol 12(10): e1005097. [read online] [download pdf]

Watch:

1. E. Bruna: How To Name Your Files

2. E. Bruna: How to Organize Your Files

3. E. Bruna: How to Store and Backup Your Files

Week 3: Data structure & format of data and datasets

Read

1. Tesi, W. 2020. An Outdated Version of Excel Led the U.K. to Undercount COVID-19 Cases. Slate. [read online] [download pdf]

2. Stolberg et al. 2020. CDC Test Counting Error Leaves Epidemiologists 'Really Baffled'. NY Times. [read online] [download pdf]

3. Broman, K. W., & Woo, K. H. (2018). Data organization in spreadsheets. The American Statistician, 72(1), 2-10. [read online] [download pdf]

4. Johnson, B. D., Dunlap, E., & Benoit, E. (2010). Organizing "mountains of words" for data analysis, both qualitative and quantitative. Substance Use & Misuse, 45(5), 648-70. [read online] [download pdf]

Week 4: Reproducible data (re)organization

Read:

1. Laskowski, 2020. What to do when you don't trust your data anymore. [read online] [download pdf]

2. Pennisi, E. 2020. Spider biologist denies suspicions of widespread data fraud in his animal personality research. Science. [read online] [download pdf]

3. Alston, J. M., and Rick, J. A.. 2020. A Beginner's Guide to Conducting Reproducible Research. Bull Ecol Soc Am 00(00):e01801. [read online] [download pdf]

4. Wilson G, Bryan J, Cranston K, Kitzes J, Nederbragt L, Teal TK (2017) Good enough practices in scientific computing. PLoS Comput Biol 13(6): e1005510. [read online] [download pdf]

Watch:

1. A Data Sharing and Management Snafu in 3 Short Acts, by the NYU Health Sciences Library

Week 5: QA/QC 1: Data Entry & Validation

Read:

1. Kamentez, A. 2018. The School Shootings that weren't.NPR [read online] [download pdf]

2. Lincoln, Matthew D. 2018. "Best Practices for Using Google Sheets in Your Data Project." [read online] [download pdf]

Week 6: QA/QC 2: Correction & Synthesis with Open Refine

No assigned reading

Week 7: QA/QC 3: Finding Errors with Data Visualization

Read:

1. Rougier NP, Droettboom M, Bourne PE (2014) Ten Simple Rules for Better Figures. PLoS Comput Biol 10(9): e1003833. [read online] or [download pdf]

Watch:

1. Tommy McCall: The simple genius of a good graphic

2. R. Luke DuBois: Insightful human portraits made from data

Week 8: Metadata, Codebooks; Data Sharing, Reuse, & Archives

Read (Metadata):

1. Michener, W.K., et al . 1997. Non-geospatial metadata for the ecological sciences. Ecological Applications 7: 330–342. [read online] [download pdf]

2. Pp 446-450 in Bernard, H.R. and Bernard, H.R., 2013. Social research methods: Qualitative and quantitative approaches. Sage. [[download pdf]]

Read (Data Sharing):

1. Alexander, S.M., Jones, K., Bennett, N.J. et al. Qualitative data sharing and synthesis for sustainability science. Nat Sustain 3, 81–88 (2020). [read online] [download pdf]

2. Renaut, S. et al. 2018. Management, Archiving, and Sharing for Biologists and the Role of Research Institutions in the Technology-Oriented Age. *BioScience* 68(6)400–411 [read online] [download pdf]

3. Duke, Clifford S., and John H. Porter. 2013. "The ethics of data sharing and reuse in biology." BioScience 63(6): 483-489. [read online] [download pdf]

4. Tenopir, C., et al. (2015). Changes in data sharing and data reuse practices and perceptions among scientists worldwide. PloS one, 10(8), e0134826. [read online] [download pdf]

5. Mauthner, NS, & O. Parry O. 2009. Qualitative data preservation and sharing in the social sciences: On whose philosophical terms?. Australian J of Social Issues 44(3):291-307. [read online] [download pdf]

Week 9: Data Management Plans

Read

1. Michener, W. K. (2015). Ten simple rules for creating a good data management plan. PLoS Computational Biology, 11(10), e1004525. [read online] or [download pdf]

2. Goodman A, et al. (2014) Ten Simple Rules for the Care and Feeding of Scientific Data. PLoS Comput Biol 10(4): e1003542. [read online] or [download pdf]. this paper connects what we've done with what's coming up next

Watch

1. Content of a DMP" (by RWTH Aachen University)

Week 10: Spring Break

No class session due to Spring Break, No assigned reading

Week 11: Efficient data collection

Read:

1. Redman, T. 2016. Bad Data Costs the U.S. $3 Trillion Per Year. Harvard Business Review. [read online] [download pdf]

Watch

1. Atul Gawande's TED Talk: The Importance & Value of the CHECK LIST.

2. Animated summary of The Checklist Manifesto.

3. Jess Stratton's LinkedIn Learning Video Overview: "Use Google Forms to Create Surveys" (7 min.). Watch it here; must be on UF computer or use UF proxy.

4. OPTIONAL: The more advanced, multi-video "Google Forms Essential Training Course". These short (1-3 min) videos explain each step in more detail. It's great, and only 39 min long from start to finish. Watch here; must be on UF computer or use UF proxy.

Week 12: Transcription & Translation

Read:

1. Bakker, Rebecca. "Transcription Tools and Software" (2017). Works of the FIU Libraries. 62. [read online] [download pdf]

2. Watch "The Text Wash team discusses text anonymization" [link]; related blog post [link]

Watch

1. Dr. Jarek Kriukow: How to transcribe interviews - Part 1: "naturalism" and "denaturalism"

2. Dr. Jarek Kriukow: How to transcribe interviews - Part 2: Which approach to use?

Week 13: Paperless data collection.

Read:

1. Aanensen DM, Huntley DM, Feil EJ, al-Own F, Spratt BG (2009) EpiCollect: Linking Smartphones to Web Applications for Epidemiology, Ecology and Community Data Collection. PLoS ONE 4(9): e6968. [read online] [download pdf]

2. Moylan, CA et al. 2013. Increasingly mobile: How new technologies can enhance qualitative research. Qualitative social work: research and practice, 14(1):36-47. [read online] [download pdf]

3. Teacher, Amber G. F. et al. Smartphones in ecology and evolution: a guide for the app-rehensive. Ecology and Evolution 3(16):5268– 5278 [read online] [download pdf]

Watch

1. E. Bruna: Paperless Data Collection

2. Great EpiCollect tutorials by UConn's David Dickson

a. Intro to EpiCollect 5
b. Creating an EpiCollect PROJECT
c. Creating a FORM for your Epicollect Project
d. Collecting data in EpiCollect with your phone or tablet

Week 14: Automated data extraction

Read:

1. Drinkwater, R. E., Cubey, R. W., & Haston, E. M. (2014). The use of Optical Character Recognition (OCR) in the digitization of herbarium specimen labels. PhytoKeys, (38), 15-30. [read online] [download pdf]

2. Joo, Jungseock and Zachary C. Steinert-Threlkeld. 2019. Image as data: Automated visual content analysis for social science. [read online] [download pdf]

Week 15: Readings Days

No class session due to Reading days, No assigned reading

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8. Academic Policies and Resources

Academic policies for this course are consistent with university policies. See https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/ for policies regarding Attendance and Makeup Work, Academic Accommodations, Grading, Course Evaluation and Feedback, the Honesty Policy, In-Class Recording, and Academic Resources.

Campus Health and Wellness Resources

Visit https://one.uf.edu/whole-gator/topics for resources that are designed to help you thrive physically, mentally, and emotionally at UF. *Please contact UMatterWeCare for additional and immediate support.*

Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Class Discussions and Group Work

In this class we may explore some challenging, important problems and increase our understanding of different perspectives and approaches for addressing them. These conversations may not always be easy; we sometimes will make mistakes in both how we communicate our perspective and what we hear other say. There may be times when we need patience, courage, imagination, and of course mutual respect to engage our texts, classmates, instructors, guests, and our own ideas and experiences. Disrespectful or disruptive behavior will not be tolerated. And always remember that as scholars we must employ critical thinking, rely on data, and cite verifiable sources and experts to interrogate all assigned readings and subject matter in this course as a means of determining if we agree with classmates and instructors. No lesson is intended to espouse, promote, advance, inculcate, or compel a particular feeling, perception, viewpoint or belief.

8.1. Evaluation

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online. Students can complete evaluations in three ways: (1) The email they receive from GatorEvals, (2) Their Canvas course menu under GatorEvals, (3) The central portal at https://my-ufl.bluera.com. Guidance on how to provide constructive feedback is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/.

Part I.

Course Topics

9. Naming Conventions

9.1. Introduction

The first (and easiest) way to get started organizing your data is to use a simple, clear, and consistent system for naming your files (i.e., a "naming convention"). Using a well-thought out naming convention has a number of important benefits:

- Files will be easier to find
- You won't have to open files to see what they are
- Files are easier to sort
- Files are easier to share with collaborators (and for collaborators to use)
- It helps prevent accidentally overwriting or deleting files

9.2. How to name your files

Your file names should:

1. **Tell you about the contents of the file and allow you to uniquely identify it.** For instance (compare `'data 1998'` vs `'survey data'` vs `'survey data 1998'` vs `'small mammal survey data 1998'`). You can use things like the an acronym for the project, abbreviation for the study, location where collected, the name of the investigator, the year or month when collected, the type of data type, etc.
2. **Be as short as possible.** Many database systems (like MS One Drive) have limits on file name length. Use a max of 25 characters, and aim for less than that.
3. **Avoid using special characters and spaces** Don't use characters like `$ % ^ & # | :`. This is really important because software, automated data processing tools, web browsers, and computer operating systems use spaces and special characters for dividing up (parsing) strings of text, as well as other processes.
4. **Start with letters, never numbers.** Using numbers at the start of file names can lead to problems when they are read into stats software and other programs. Instead of `'2020_survey_responses'` use `'survey_responses_2020'`
5. **Help you quickly sort files chronologically or numerically**. Dates are very useful in file names because they can help identify the most recent files easily. You can *not* rely on the file date associated with the file on the computer! However, it is really important to use a clear format for dates, especially if collaborating internationally. I learned the hard way that not everyone reads `'census_data_3-4-2018'` the same way I do, so I started using `'census_data_4march2018'`. However, even

this isn't ideal because of how it sorts things in a folder. Now I now use this format: `'file_name_YYYYMMDD'` (`'census_data_20180403'`). You can also use `file_name_YYYY-MM-DD`. This is based on the international format for dates and times (ISO 8601). In addition to the benefit of sorting more easily, you'll never have to use the the dreaded `'_____.final.docx'` and `'_____.actual_final.docx'` in your file names ever again…you'll always know which is the last one. If the files don't require a date, but you want to maintain a sequence, use leading zeros to maintain sort order. `7` and `701` don't sort the way you expect them to, so use `007` and `701`.

6. **Use a consistent method of dealing with spaces and letter case.** . For a the spreadsheet with responses to a survey by 5 different people, don't use `'survey responses Florida'`. Instead use (i) `'survey_responses_florida'` or (ii) `'SurveyResponsesFlorida'`. Try to avoid mixing the two (e.g., `'survey_responses_Florida'`). By the way, (i) is known as `pothole` and (ii) is known as `camel`. Personally, I prefer pothole name is better because of it is easier to read and you never to remember what words get capitalized (because none of the do). Use underscore (`_`) or dashes (`-`) to separate meaningful parts of file names.

Once you have a system, write it down, print it out, and put it up somewhere you can easily refer to it. Remember -whenever possible make the names simple, informative, and unique. If you collected survey data and behavioral data in 2017, don't call them both `data_2017` even if they are in different folders. Call one `'behavior_obs_2017'` and the other `'survey_responses_2017'`).

Simple names also help you avoid problems with "system and software portability" - you might be working on a Mac but your collaborators are working on Linux or Windows, and the software made for each can read file names in different ways. File names that are simple, lower-case, and have no special characters are less software- and platform-dependent.

9.3. Tools & Resources

9.3.1. Bulk Renaming of Files

- Adobe Bridge
- Bulk Rename (Windows)
- Rename Master (Windows)
- File Renamer Basic (Windows, Free and Paid)
- Ant Renamer
- Renamer 5 (macOS)
- Renamer Lite
- PSRenamer
- Batch Renaming of Files in Windows: Post 1, Post 2
- Batch Renaming of Files in MacOS: Post 1, Post 2

9.3.2. Best Practices - Other Institutions

- Smithsonian Library

- Michigan Library
- Stanford Library
- Smithsonian

Part II.

Appendix 1: Useful Resources

10. Useful resources for Data Collection & Managment

10.1. R Programming

10.1.1. Essential

1. Hadley Wickham wrote a book on using the tidyverse and the online version is FREE. This is a phenomenal resource on using R to import, tidy, and visualize data.
2. Posit Cheat Sheets: help with commands for using the different `tidyverse` packages, RStudio shortcuts and tricks, help with R commands, and more. You definitely want the ones for Data Import, Work with Strings, Factors, Data Transformation, and Base R.
3. Where and How to ask for help
   - Hadley Wickham's advice on how to write a good reproducible example for getting help with R
   - how to post good questions on StackOverflow
   - The UF R-users listserv is *very* user friendly and a great place to post requests for help.

10.1.2. Tutorials and Books

1. R Essential Training: Wrangling and Visualizing Data
2. Software Carpentry: Using RStudio for Project Organization & Management
3. Swirl
4. R Bootcamp
5. Kieran Healy's *Data Visualization: a practical introduction* is my favorite introductory (yet super-comprehensive) book on data visualization with R. If you scroll down to the bottom of the page you can download the datasets and code used to make the figures in the book, which makes life much easier.
6. So. Many. Resources.
7. ROpenSci: tools for accessing, manipulating, and visualizing open data

8. How to clean messy data in R
9. The Ultimate Guide to Data Cleaning
10. Learning R
    Swirl

11. Specific Problems in Data Cleaning and Managemnt

1. Handling dates and times in R
2. Text Mining: `tidytext` package
3. Working with Qualtrics survey data with the `qualtRics` package
4. Optical Character Recognition (OCR): extract text from images: `tesseract` package
5. Extract text & metadata from pdf files: `pdftools` package
6. Image processing in R: the `magick` package

11.0.1. Advanced R Packages

1. `DataCurator` package: 'a simple desktop data editor to help describe, validate and share usable open data'.
2. RegExr: online tool to learn, build, & test Regular Expressions (RegEx / RegExp)
3. janitor (cleanup of file names, etc.)
4. `knitr` overview: reproducible documents with R
5. qualtRics
   ## Discipline-specific Resources
6. `historydata` package: Sample data sets for historians learning R. They include population, institutional, religious, military, and prosopographical data suitable for mapping, quantitative analysis, and network analysis.
7. *The Programming Historian* Website: wide range of topics, from text analysis to OpenRefine

11.1. Slide Presentations in R

1. Make slide presentations with R

11.2. Data Archives

Qualitative Data Repository

11.3. Text Extraction and Organization

Plan for extraction and organization

11.4. Form Design

Best Practice for Form Design

11.5. Data Security

UF Office of Information Security and Compliance
Cyber Safeguards for UF
UF IRB
UF Data Classification Policy
UF Office of Information Security and Compliance

11.6. Platforms for Organization & Collaboration

Open Science Framework